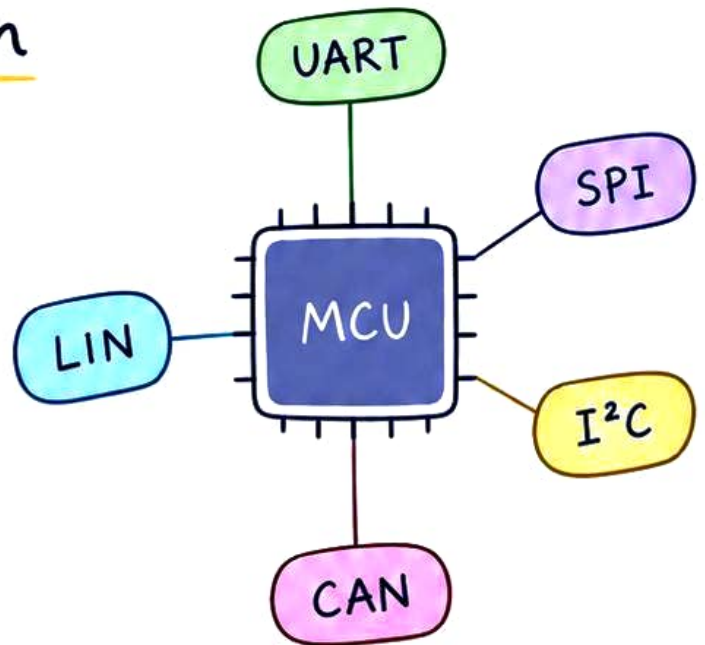


Communication & Networking

- UART Communication
- SPI Communication
- I²C Communication
- CAN Bus
- LIN Bus
- Wi-Fi
- Bluetooth
- Ethernet
- GPS/GNSS

Different protocols...
Same goal



Better Communication...
Smarter Systems



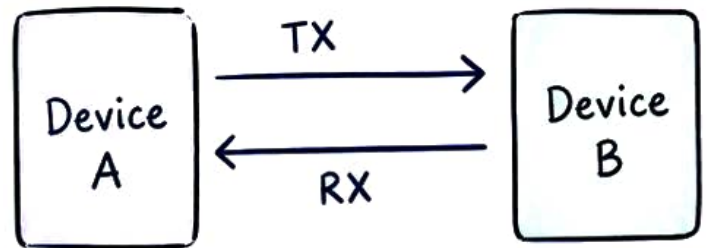
UART

Communication

(Universal Asynchronous Receiver Transmitter)

1. What is UART ?

- A serial communication protocol used to transmit data between two devices.
- Works in an asynchronous manner (no shared clock).
- Uses 2 main lines:
 - TX (Transmit) → Sends data
 - RX (Receive) → Receives data



(Point to Point Communication)

2. Key Features

- Simple and easy to use.
- No clock line required.
- Full-duplex (send and receive at the same time).
- Widely used in microcontrollers, sensors, GPS, modules, etc.

3. Frame Format



← 1 frame →

- Start bit : 0
- Data bits : 5 to 9 (usually 8)
- Parity bit : optional (error check)
- Stop bit : 1 or 2 (end of frame)

4. Common Baud Rates

9600, 19200, 38400, 115200, etc.



UART is simple, reliable and still one of the most used communication methods in embedded systems.

SPI

Communication

(Serial Peripheral Interface)

Fast
Simple
Full Duplex

1. What is SPI ?

- A synchronous serial communication protocol used to exchange data between a master and one or more slave devices.
- Works in full-duplex mode (send and receive at the same time).
- Uses 4 main lines :

MOSI → Master Out Slave In
(data from master to slave)

MISO → Master In Slave Out
(data from slave to master)

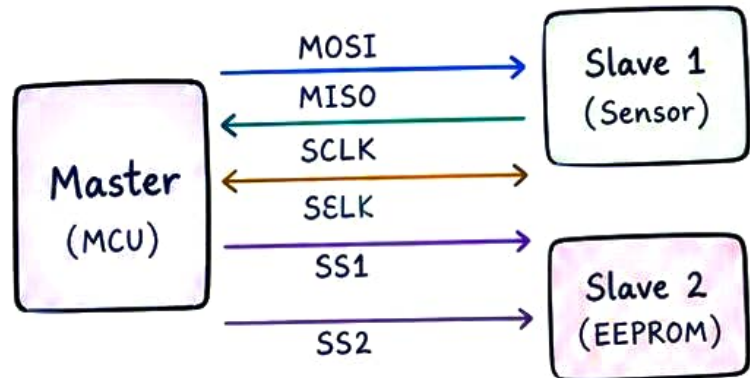
SCLK → Serial Clock
(generated by master)

SS/CS → Slave Select
(active low)

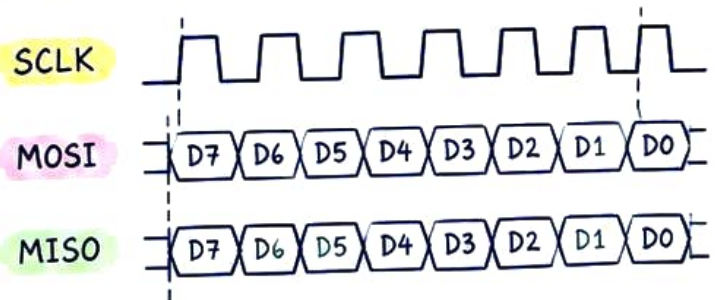
4. Key Features

- High speed (up to several Mbps).
- Simple hardware.
- Supports multiple slaves (each with its own CS line).
- Full-duplex communication.
- Widely used in sensors, memory, displays, and microcontrollers.

2. System Diagram



3. Data Transfer (Timing)



Data is sampled on the rising/falling edge of SCLK (depends on device).



SPI is fast, reliable and widely used for short-range communication between devices.

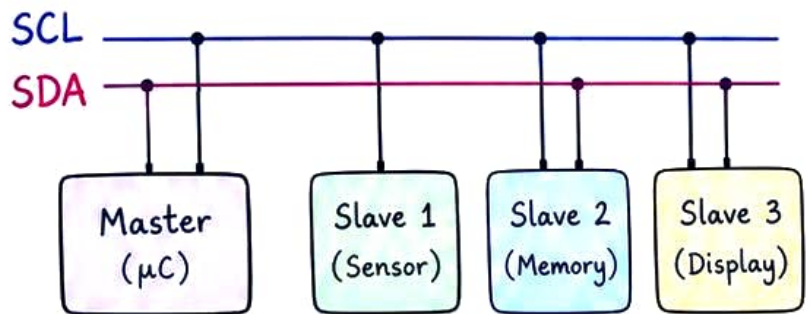
I²C Communication

(Inter-Integrated Circuit)

Simple
Two Wire
Protocol

1. What is I²C ?

- A synchronous, multi-master, multi-slave serial communication protocol.
- Uses only two wires :
 - SCL → Serial Clock
 - SDA → Serial Data
- Allows multiple devices to share the same bus.



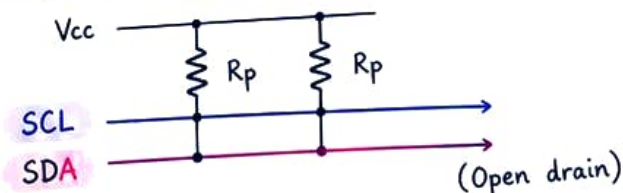
All devices share the same SCL and SDA lines.

2. Key Features

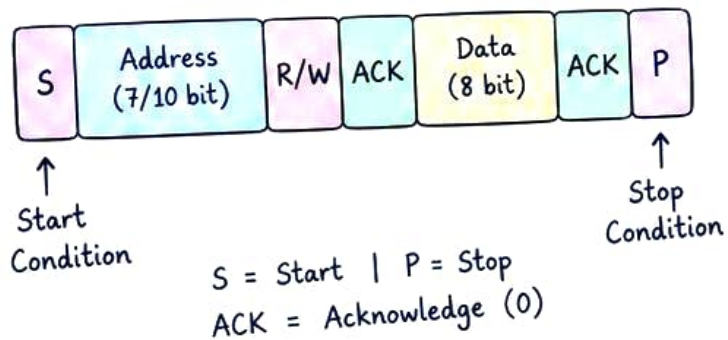
- Simple and low cost.
- Supports multiple devices (addressing).
- Supports multi-master (bus arbitration).
- Data can be up to 100 kbps (Standard mode), 400 kbps (Fast mode), 1 Mbps (Fast-mode Plus).
- 7-bit or 10-bit addressing.

4. Key Signals

- SCL : Clock signal (generated by master).
- SDA : Data signal (open-drain).
- Both lines have pull-up resistors.



3. Frame Format



I²C is widely used in sensors, EEPROM, RTC, displays and many embedded systems.

CAN

Communication

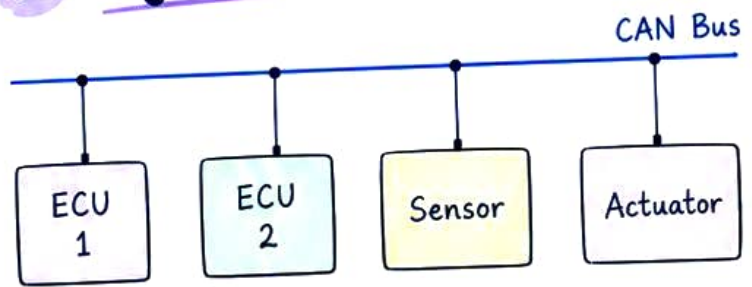
(Controller Area Network)

Reliable
Fast
Robust

1. What is CAN?

- A serial communication protocol used to connect multiple devices in a network (mostly in vehicles).
- Designed for real-time and high reliability communication.
- Developed by **Bosch** in 1986.

2. System Diagram

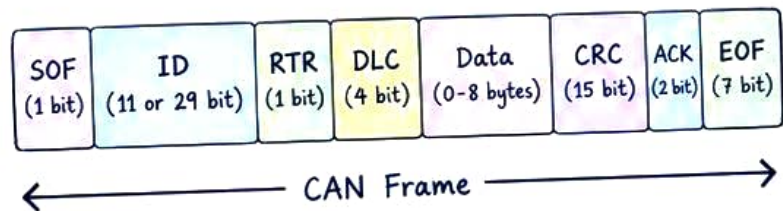


(All devices are connected in parallel on the same bus)

3. Key Features

- Multi-master (multiple devices can transmit).
- Message-based communication.
- High noise immunity.
- Error detection and fault confinement.
- Supports real-time communication.
- Up to 1 Mbps (CAN FD supports higher speeds).

4. Frame Format



ID : Identifies the message (priority also).

RTR : 0 = data frame, 1 = remote frame.

DLC : Number of data bytes.

5. CAN vs Other Protocols

Feature	CAN	UART / SPI / I ² C
Speed	Up to 1 Mbps	Lower
Topology	Multi-device (bus)	Point-to-point (or simple bus)
Error Handling	Built-in	Limited / None
Use Case	Automotive, industrial	General purpose



CAN is a robust and efficient protocol for reliable communication in critical systems like vehicles, robotics and industrial automation.

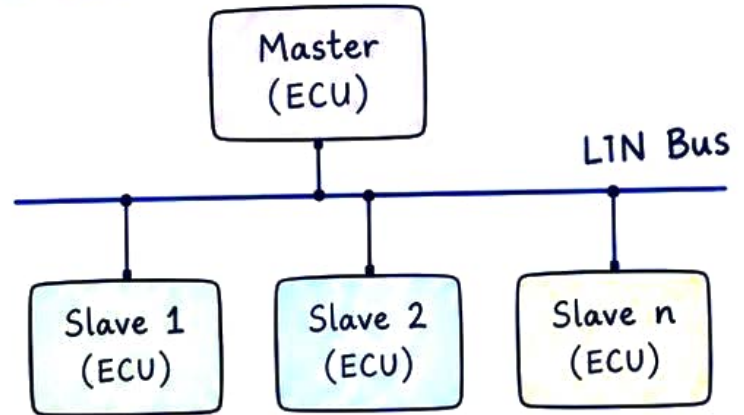
LIN Bus

(Local Interconnect Network)

1. What is LIN?

- A low-cost, single-wire communication protocol.
- Used mainly in automotive for simple, low-speed communication between smart devices.
- Often works with CAN (CAN for high speed, LIN for low cost tasks).

2. System Diagram



One master, multiple slaves.

3. Key Features

- Single wire (data line).
- Low cost & simple design.
- Low data rate (up to 20 kbps).
- Master-slave communication.
- Built-in error checking (checksum).

4. Frame Format



← LIN Frame →

- Break : indicates start of frame
Sync : helps with synchronization
PID : identifies the message
Checksum : error detection

5. Applications

- Door control
- Window lift
- Lighting systems
- Climate control (HVAC)
- Seat adjustment



Used in cars!



LIN is simple, low-cost and perfect for small, low-speed communication needs in vehicles.

Ethernet Basics

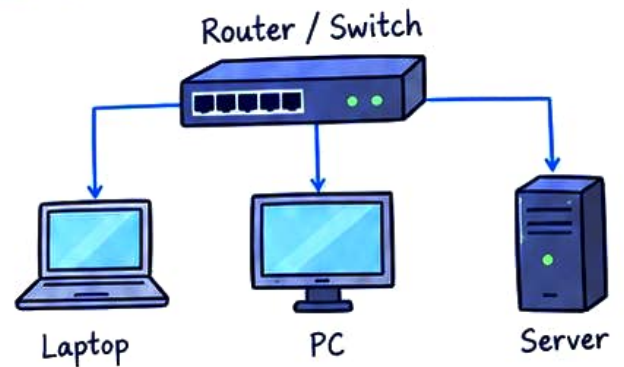
(Local Area Network)

1. What is Ethernet?

- Ethernet is a wired network technology used to connect devices in a LAN (Local Area Network).
- It provides fast, reliable and standardised communication between devices.
- Widely used in homes, offices, industries and data centres.



2. How It Works?

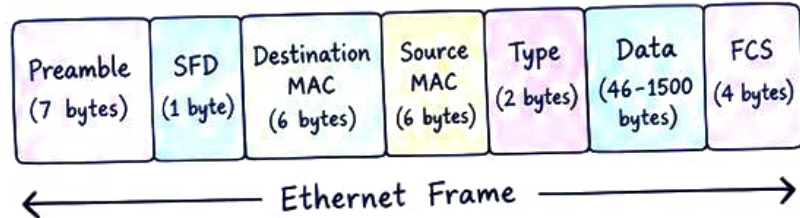


Devices are connected through Ethernet cables and communicate using standard protocols (like TCP/IP).

3. Key Features

- High speed (10 Mbps to 400 Gbps).
- Reliable and stable connection.
- Uses standard cables and connectors (RJ45).
- Supports full-duplex communication (send & receive at the same time).
- Scalable (works in small and large networks).

4. Ethernet Frame Format



MAC : Media Access Control address

SFD : Start Frame Delimiter

FCS : Frame Check Sequence (error check)

5. Common Ethernet Standards

- 10BASE-T → 10 Mbps (early standard)
- 100BASE-TX → 100 Mbps (Fast Ethernet)
- 1000BASE-T → 1 Gbps (Gigabit Ethernet)
- 10GBASE-T → 10 Gbps (10 Gigabit Ethernet)

Ethernet makes modern communication faster, smarter and more connected!



Ethernet is the backbone of modern networks — from your home Wi-Fi router to global internet infrastructure.

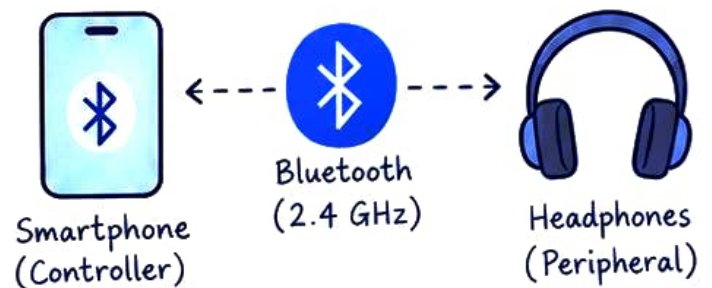
Bluetooth

(Short Range Wireless Communication)

1. What is Bluetooth?

- A wireless technology used to connect devices over short distances.
- Works in the 2.4 GHz ISM band (same as Wi-Fi).
- Enables data, audio and control communication between devices.

2. System Diagram



Devices communicate directly or through a central device.

3. Key Features

- Operates in 2.4 GHz band (79 channels).
- Low power consumption.
- Supports multiple data rates (1 Mbps – 3 Mbps).
- Pairing and security (encryption).
- Works over short range (~10 meters, class 2).

4. Bluetooth Versions

Version	Data Rate (max)	Range (approx.)
4.0 / 4.2	1 Mbps	~10 m
5.0	2 Mbps	~10 – 20 m
5.3	2 Mbps	~10 – 20 m

(Later versions improve range, speed and power efficiency.)

5. Applications

- Wireless headphones and speakers.
- Fitness trackers and smart watches.
- Keyboards, mice and game controllers.
- IoT devices and home automation.
- Vehicle hands-free systems.



Bluetooth is simple, low power and perfect for everyday short-range connections.

Wi-Fi

Topic: Wi-Fi

(Wireless Local Area Network)

1. What is Wi-Fi?

- Wi-Fi is a wireless networking technology used to connect devices to a local network (LAN) and internet.
- Uses radio waves instead of cables.
- Based on IEEE 802.11 standards.
- Commonly used in homes, offices, schools and public places.

2. System Diagram



Devices connect to the router (access point) using Wi-Fi.

3. Key Features

- High data rates (up to hundreds of Mbps).
- Works in 2.4 GHz and 5 GHz bands.
- Supports multiple devices.
- Secure with encryption (WPA2/WPA3).
- Mobile and flexible (roaming).

4. Wi-Fi Standards

Standard	Frequency Band	Max Speed (approx.)
802.11b	2.4 GHz	11 Mbps
802.11g	2.4 GHz	54 Mbps
802.11n	2.4 / 5 GHz	~150 Mbps
802.11ac	5 GHz	~1 Gbps
802.11ax (Wi-Fi 6)	2.4 / 5 GHz	~9.6 Gbps

(Later versions are faster and more efficient.)

5. Applications

- Home internet (phones, laptops, smart TVs).
- Office and enterprise networks.
- Smart devices (IoT).
- Public hotspots (cafés, airports, etc.).
- Industrial and automation systems.



Wi-Fi gives us wireless freedom — connecting devices, people and ideas anywhere!

GPS / GNSS

(Global Positioning System / Global Navigation Satellite System)

1. What is GPS/GNSS?

- GNSS is a satellite-based navigation system that provides position, velocity and time (PVT) information.
- GPS is the U.S. system, GNSS includes other systems too (like Galileo, GLONASS, BeiDou, QZSS).
- Used in vehicles, drones, mapping, tracking, smartphones and more.

3. Key Features

- Provides accurate position (latitude, longitude, altitude).
- Uses multiple satellites (4 or more).
- Works in all weather conditions.
- Global coverage (land, sea, air).
- Typical accuracy: 2-10 meters (standard GPS).

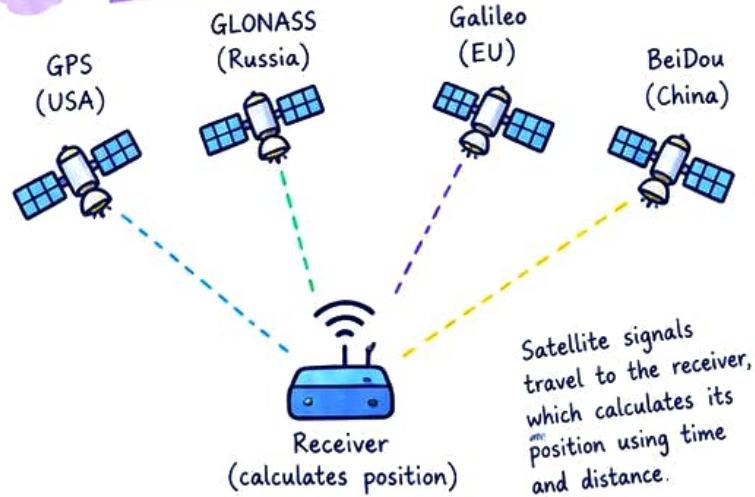
5. Applications

- Navigation (cars, ships, aircraft).
- Location tracking (phones, wearables).
- Mapping & surveying.
- Farming & agriculture.
- Military and disaster management.



Find your way, anywhere!

2. How It Works?



4. GNSS Systems

System	Operator	Satellites (approx.)	Region
GPS	USA	31	Global
GLONASS	Russia	24	Global
Galileo	EU	24	Global
BeiDou	China	35+	Global
QZSS	Japan	4	Asia-Pacific

(More satellites = better availability & accuracy)

6. Important Terms

- Latitude : North/South position.
- Longitude : East/West position.
- Altitude : Height above sea level.
- Ephemeris : Satellite orbit data.
- Receiver : Device that gets and processes the signals.



Satellites connect the world!

CAN vs LIN vs Ethernet

CAN

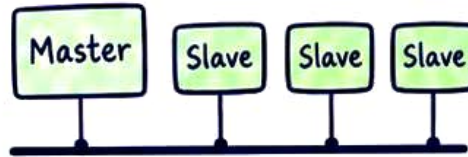
(Controller Area Network)



- Used in modern vehicles (engine, ABS, airbags etc.)
- Multi-master bus
- High speed & reliable
- Error detection (built-in)
- Uses differential signals (CAN High / CAN Low)

LIN

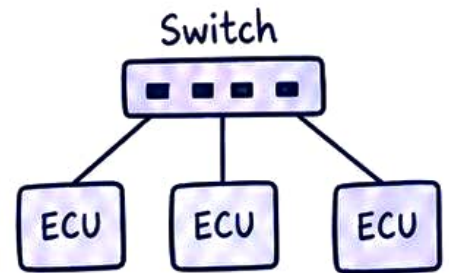
(Local Interconnect Network)



- Used in simple systems (windows, seats, lights etc.)
- Single master
- Low cost
- Slower speed
- No error correction (only detection)
- Uses single wire

Ethernet

(IEEE 802.3)



- Used in infotainment, telematics & large data transfer
- High speed (100 Mbps - Gbps)
- Full duplex
- More complex and expensive



In short:

CAN = Reliable | LIN = Simple & Cheap | Ethernet = Fast & High Data